

CONTACT

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TECHNICAL SKILLS

Languages

JavaScript · TypeScript · Java ·

Python · R

Frameworks & libraries

Node.js · Express.js · React.js ·

Next.js · Angular · Spring · Spark

Databases

MySQL · PostgreSQL · PostGIS ·

MongoDB · Cassandra

DevOps & deployment

Docker · Docker Compose · nginx ·

Git

Testing

Jest · Vitest · Supertest · Testing

Library

Automation

n8n

LANGUAGES

Spanish Native

English B2 — Cambridge B2 First

STRENGTHS

- Analytical thinking
- Time management
- Continuous learning

AVAILABILITY

Immediate. On-site, hybrid or remote, and open to relocating anywhere in Spain.

José Carlos Ruiz Sánchez

Full-Stack Developer

Node.js · Express.js · React · TypeScript · PostgreSQL · PostGIS · Docker

Full-stack developer with a degree in Computer Engineering and a Master's in progress. During my internship at the Applied Computing Group (University of Almería) I led most of the development of an interactive map of the devices connected to the campus Wi-Fi, refreshed every two minutes by an automated ingestion pipeline. On my own initiative I built Tokimori, a full-stack app running six Node.js/TypeScript microservices behind a React 19 front end. Portfolio and project write-ups: jrs407.github.io/web-personal

EXPERIENCE

Full-Stack Developer — Extracurricular Internship

Feb 2026 – Aug 2026

Applied Computing Group (ACG) · University of Almería · Almería, Spain

- Built a **REST API** in Node.js with Express.js, covered by Jest test suites.
- Designed the **geospatial data model** in PostgreSQL/PostGIS and an **n8n ingestion pipeline** handling ~3,000 rows every 2 minutes, with the data encrypted to preserve anonymity.
- Refactored and extended an inherited **Next.js** front end: map layers for hexagon binning, heatmaps, building occupancy, routes between buildings and floor-level detail, plus a historical mode with time playback of past data.
- Deployed with **Docker** behind an **nginx** reverse proxy.

PROJECTS

Tokimori

Feb 2026 – Aug 2026

Full-stack web application · personal project

- **Six microservices** in Node.js, Express 5 and TypeScript behind an nginx **API gateway**, with MySQL, Docker Compose (separate production and development configurations) and a React front end.
- **Visual editor built on the HTML5 Canvas API with no external libraries:** shapes, images, text, rotation, layers and undo/redo.
- **JWT authentication** over secure cookies with bcrypt, helmet and rate limiting. Tested with Vitest and Testing Library on the front end, Jest and Supertest on the back end.

Burnout Predictor

May 2026

Machine learning · Python & Spark

- Python model that predicts the risk of burnout from a set of supplied criteria, deployed across a **Spark** node cluster to speed up data processing.

Auscultify

Feb 2026

Bachelor's thesis · graded 9.4/10 (Spanish scale, out of 10)

- Complete application in React.js, Node.js and Python built on a **microservice architecture** with AI-based recommendation algorithms, built to teach medical students to recognise stethoscope recordings.

EDUCATION

MSc in Computer Engineering

Oct 2025 – present

University of Almería

Big Data specialisation. Structured to be compatible with full-time employment.

BSc in Computer Engineering

Sep 2021 – Feb 2026

University of Almería

Double specialisation in Software Engineering and Information Systems.